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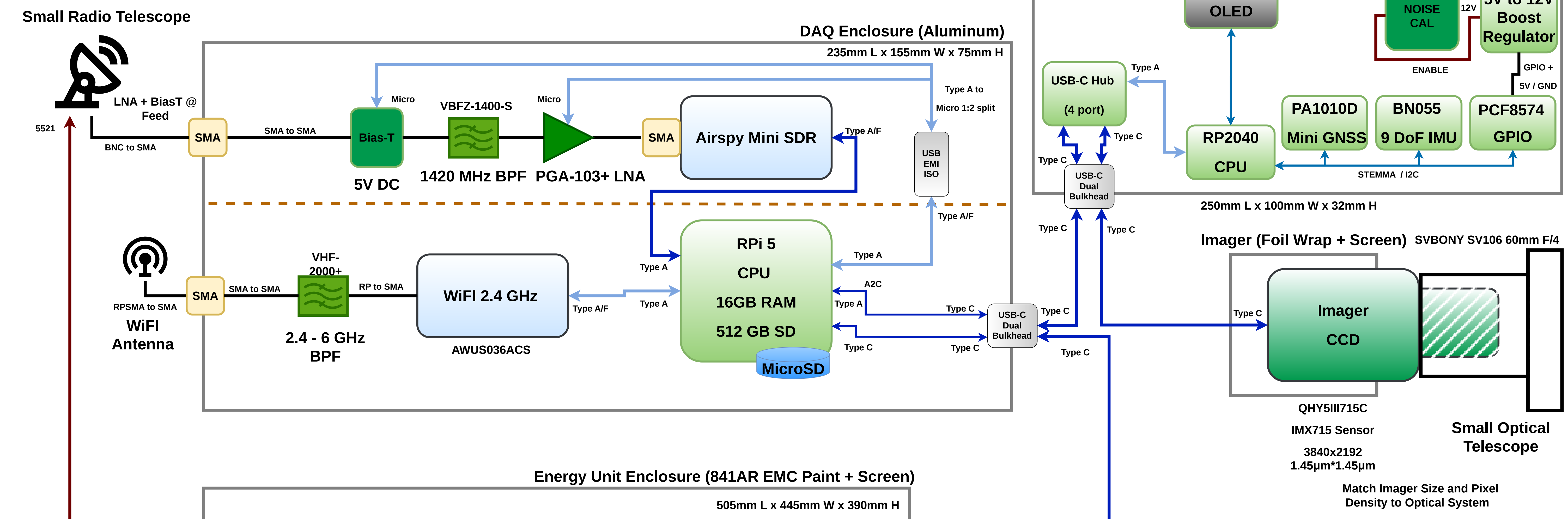
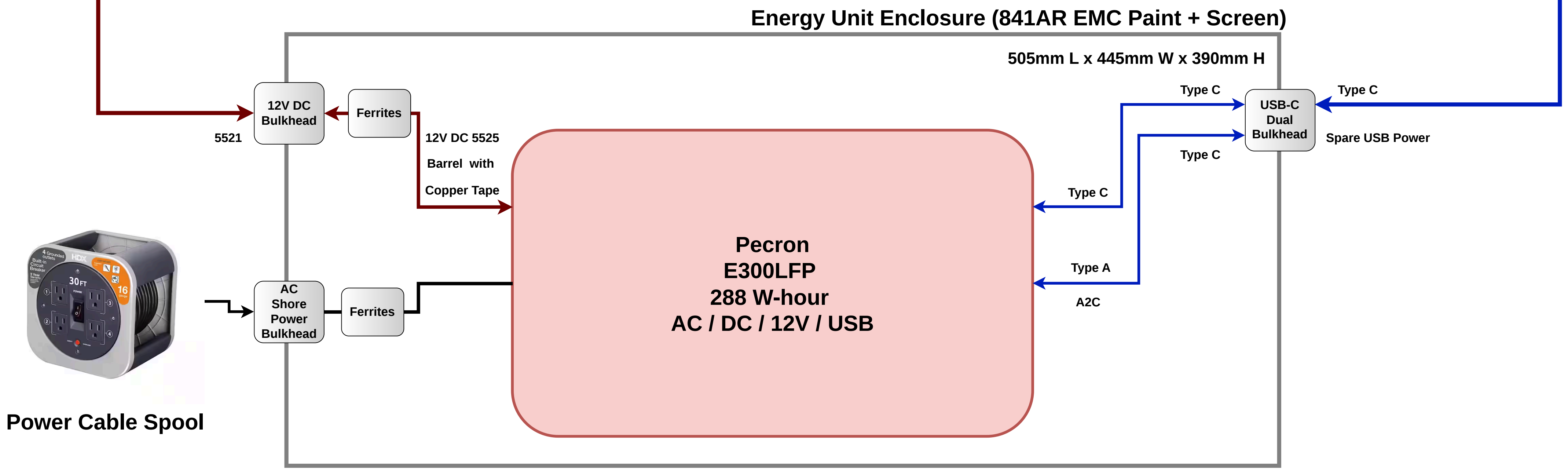
ASTRA (Automated Small Telescope for Radio Astronomy)

Architectural Design

Revision 1.0 - FDL





ASTRA integrates a small radio telescope, an optical wide-field imaging system, a position and orientation sensor, calibrator, and an astronomical goto mount. An on board computer enables automation and control of the system. Combined with a browser based interface to student laptops the platform can be used for collection and visualization of radio and optical astronomy data. This enables active learning for science and radio education through real world measurement.

The system is relatively inexpensive and suitable for construction and use by a science program or astronomy club. ASTRA can be used as an adjunct for lesson plans related to the electromagnetic spectrum, chemistry and physics, and radio and optical astronomy. More advanced 'mini projects' are also possible for small groups of enthusiastic students who are willing to put in more work observing and analyzing the resulting data.

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Program: Adaptive Radio Science (ARS)			
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